

Assessing the Feasibility of Behind-the-Meter Battery Storage Systems for Tariff Arbitrage in Uganda

Tunmise Raji

Fhazhil Wamalwa

Lefu Maqelepo

Nathaniel J. Williams

Golisano Institute for Sustainability

Rochester Institute of Technology

Rochester, NY USA

Abstract—The challenge of providing affordable electricity to users in sub-Saharan Africa persists. This study examines the feasibility of behind-the-meter battery energy storage systems (BESS) for tariff arbitrage. We utilize the time-of-use tariff from Umeme, a distribution utility in Uganda, to develop an optimization and economic model that compares electricity costs with and without a BESS for commercial use. Our results reveal that while storage systems can significantly decrease electricity costs through load shifting and peak shaving, the high initial investment required to install the BESS prevents them from being cost-effective. To explore alternative options, we examine a scenario with second-life BESS, which has lower capital costs and shorter lifetimes than a new BESS. Our results demonstrate that in this scenario, the payback of the system could be as low as 2 years. This study provides insight into potential solutions to reduce electricity costs for grid-connected users in sub-Saharan Africa.

Index Terms—Tariff, Uganda, Storage, Optimization

I. INTRODUCTION

The affordability of electricity remains a critical challenge for both residential and commercial consumers in sub-Saharan Africa (SSA). A 2016 World Bank study found that commercial and industrial customers in SSA pay as high as \$0.5/kWh which is significantly higher than the average rate of \$0.08/kWh paid by US industries [1], [2]. The cost of electricity remains high even though governments provide significant subsidies to utilities and utilities often do not charge cost-reflective tariffs [3]. Several reasons have been highlighted for the high cost of electricity in SSA including electricity theft, low consumption of electricity, high cost of capital, dependence on foreign fuel imports, high transmission losses, geography, and size of the market, among others [1]. The high cost of electricity can also be expected to increase in the future given that of the 39 countries evaluated in a World Bank study that examined the financial position of utilities in SSA, only two (that is, Uganda and Seychelles) were recovering their capital and operational expenses from the cash collected from electricity sales [1]. The tariff structure in one of the cost-recovering countries, Uganda, is examined in this paper.

Given the key role that access to affordable and reliable electricity played in the industrialization of developed countries in the 19th and 20th centuries [4], it is important to explore solutions to reduce the cost of electricity in SSA countries. One of the reasons for the high cost of electricity generation in the region is the increasing use of leased emergency power generation systems, which are required to reduce power outages [5]. Some of these power outages are caused by the high night-time loads that exist in most power systems in the region. Power system operators, therefore, require fast ramping generators to meet their nighttime peak. Aligning the demand for electricity with its supply by load leveling can reduce the use of these expensive backup systems. Time-varying tariffs are important instruments through which utilities signal the time-varying service delivery costs to consumers. Although most utilities in the region initially charged a flat tariff irrespective of when electricity is consumed, many utilities have begun to implement time-varying tariffs with 20 of the 38 countries in the region surveyed in a World Bank study implementing tariffs that segment the days into two or three periods [6].

The expectation of utilities when designing time-varying tariffs is that electricity users will shift their consumption from peak times to off-peak times. Research has, however, shown that time-varying tariffs have little impact on the time of consumption of electricity in microgrids in the region [7]. Although microgrids are made up of predominantly residential electricity users, a similar behavior may be expected among industrial and commercial electricity users who have inflexible loads.

In this study, we propose a behind-the-meter battery energy storage system (BESS) for tariff arbitrage in a utility with a time-of-use (TOU) tariff. Tariff arbitrage involves storing energy in a BESS when electricity from the grid is cheap and retrieving it to power loads in peak periods. This solution has the benefit of not requiring flexible loads or demand-side management strategies to allow customers to take advantage of TOU tariffs. Several studies have evaluated the feasibility

of using BESS for price arbitrage, deferring the upgrade of distribution networks, or other ancillary grid services such as frequency control in wholesale electricity markets in Great Britain [8], United States [9], Austria [10], Australia [11], among others. These studies typically involve the use of optimization models to determine the optimal energy and power capacity of battery systems, and the optimal charge and discharge schedules to maximize profitability. Fewer studies have examined the feasibility of behind-the-meter BESS for reducing electricity costs, especially on the African continent. Behind-the-meter BESS was assessed for peak shaving in an academic building with grid-connected solar systems in South Africa in [12] although the study did not consider time-varying tariffs and the main function of the BESS, in this case, was to reduce demand charges by reducing the maximum power draw. Another study also considered the feasibility of using BESS with TOU tariff in South Africa and found that the assumed days of autonomy, which further impacts the size of the storage system, has a large impact on the financial feasibility of the solution [13]. However, the study assumed an optimistic lifespan of 20 years for the lead acid (PbA) and lithium-ion batteries at 60% depth of discharge. Behind-the-meter BESS was also assessed for tariff arbitrage and to reduce demand charges in [14] although the capital expenditure required to purchase the BESS was not considered.

We address these gaps by investigating the feasibility of a behind-the-meter BESS installed in the Ugandan electricity market to reduce electricity costs through tariff arbitrage and peak shaving. Furthermore, we also consider the costs associated with the installation of the storage system to enable a comprehensive economic assessment of the solution. We investigated the feasibility of this solution under several scenarios including BESSs with different lifetimes, different unit costs, and new and second-life BESS. The following section describes the data sources used in this study, followed by a comprehensive overview of both the battery and optimization models. Subsequently, the results and the consequential significance of our discoveries are presented.

II. DATA

A. Load profile

The hourly resolved load profile for a commercial application was obtained by digitizing the load profile presented in a Uganda-focused market insight report supported by the European Union that assessed the feasibility of investments in photovoltaic power systems in the country [15] and is shown in Fig. 1.

B. Tariff Structures

Since the maximum hourly demand in the assumed load profile is around 67kW, the commercial facility will be classified under the Medium Industrial Consumers (CODE 20) category [16]. Table I provides a summary of the TOU tariff for this category approved by the Ugandan Electricity Regulatory Authority to be charged by Umeme from January to March 2023 [16]. The tariff has an energy charge component (UGX/kWh),

a power charge component also known as maximum demand charge (UGX/kW), a reactive energy charge (or reward), and a monthly charge. A maximum demand charge of 16,644 UGX/kVA/month [17] was used while the monthly charge and the reactive energy charge were not considered since they are not likely to be significantly impacted by the BESS. Since the analysis is limited to a day, we convert the monthly maximum demand charge to a daily maximum demand charge by dividing it by a factor of 30. Lastly, a power factor of 0.8 was used to convert the base of the maximum demand charge from kVA to kW.

TABLE I: Electricity rate

Name	Time	Base Tariff (UGX/kWh)
Peak	6 PM - 12 AM	646.9
Shoulder	6 AM - 6 PM	476.8
Off-peak	12 AM - 6 AM	255.4

Perfect foresight of the load and electricity cost data was assumed therefore the findings may represent the best-case scenario. It is reasonable to assume that we have perfect foresight on the TOU rates since they are set once every quarter.

III. METHODOLOGY

A. Battery energy storage system model

The battery energy storage system model takes into account key lithium-ion battery characteristics such as energy capacity, power rating, charging efficiency, and discharging efficiency as detailed in Table II.

TABLE II: Notations

Parameter	Symbol	Unit	Value
<i>Battery parameter</i>			
Battery capacity	E_b	kWh	395
Power rating	P	kW	70
Energy flowing into the battery at hour i	E_i^c	kWh	
Energy flowing out of the battery at hour i	E_i^d	kWh	
Charge efficiency	η_c	-	95%
Discharge efficiency	η_d	-	95%
Depth of discharge	DoD	-	80%
State of charge at hour i	SoC_i	kWh	-
<i>Tariff parameter</i>			
Tariff at hour i	T_i	UGX/kWh	-
<i>Load parameter</i>			
Load at hour i	L_i	kWh	-
Grid import at hour i	G_i	kWh	-

Equation (1) below details how the base capacity of the battery E_b ($\approx 395kWh$) was estimated from the total peak time energy demand with the battery sized to be able to meet the peak time load. The power rating of the battery P was calculated as $\approx 70kW$ by dividing the maximum hourly demand of the battery by the discharge efficiency to ensure that the battery can power the maximum load.

$$E_b = \frac{\sum_{i=6PM}^{12AM} L_i}{DoD \times \eta_d}, \quad (1)$$

We also explore other battery capacities ranging from 50kWh to 1MWh in steps of 25kWh to assess how battery sizes affect the feasibility of the solution.

B. Optimization model

The aim of the optimization model is to determine the optimal charge and discharge schedule of the BESS, considering the state of charge (SoC) of the battery and the cost of electricity to minimize the cost of daily electrical energy. Since the electricity cost has a maximum demand charge component and an energy cost component, the objective function of the study problem is expressed as follows:

$$J = \min \left\{ \left(\sum_{i=1}^{24} (G_i \times T_i) \right) + \left(\frac{DC}{30} \times P_{max} \right) \right\}, \quad (2)$$

where DC , and P_{max} are respectively the maximum demand charge and the peak power demand. All other notations are as defined in table II. The objective function, Eq. (2), is solved subject to the following constraints:

$$SOC^{min} \leq SOC_i \leq SOC^{max}, \quad (3)$$

$$SOC_i = SOC_{i-1} + \eta_c E_i^c - E_i^d / \eta_d \quad (4)$$

$$G_i = L_i + E_{i,c} - E_{i,d} \quad (5)$$

$$P_{max} = \max\{E_i / \Delta t\} \quad ; \quad \Delta t = 1 \text{ hr} \quad (6)$$

$$E_i^c \geq 0; \quad E_i^d \geq 0; \quad SOC^{min} = (1 - DoD) \times E_b \quad (7)$$

where SOC^{min} and SOC^{max} are minimum and maximum SOC of the battery. Eq. (3) requires that at any given time i , the battery SOC_i is constrained to lie between SOC^{min} and SOC^{max} . Eq. (4) is a dynamical expression of the SOC of the battery: at any given time i , the SOC of the battery is given by the SOC at the previous time instance $i - 1$ plus the charge of the energy charged or discharged from the battery during that time instance i . According to Eq. (5), the hourly grid energy import is equal to the hourly load demand, L_i , plus the hourly battery energy charge, E_i^c , less the hourly battery energy discharge, E_i^d . The optimization was implemented with Pyomo (v6.5.0) [18] and solved with CPLEX solver.

C. Economic analysis

We assess the economic impact of the BESS solution on electricity cost with two metrics: discounted payback period and net present value (NPV) of cash flows. First, the diurnal electricity cost (EC) was calculated as shown in Eq. (8) and (9) for the scenarios with and without the BESS. The perfect foresight assumption mentioned earlier leads us to assume that the daily cost savings (that is, the difference between the EC s) will be constant throughout the lifetime of the battery. The discounted payback period DPP that informs us of how long it will take for the net present value of the savings achieved by the operation of the BESS to add up to the capital expenditure required to implement the BESS is shown in Eq. (10) where C refers to the unit cost of the BESS and r is the discount rate. The unit cost of lithium-ion BESS reported in the literature

for minigrid applications, especially in Africa, is uncertain and higher than the global industry benchmarks (\$605 vs \$209 per kWh) [19], [20]. Eq. (11) and (12) were used to calculate the NPV of the electricity cost with and without the BESS over the system's lifetime where Y refers to the lifetime of the BESS. With a depth of discharge of 80%, the cycle life of a lithium-ion battery varies between 1,900 and 3,000 cycles, which corresponds to a lifetime of approximately 5 and 8 years, respectively, assuming a cycle per day [21]. A discount rate (r) of 7% and a currency exchange rate of \$1 to UGX3700 were also assumed.

$$EC_{Without} = \left(\sum_{i=1}^N T_i \times L_i \right) + DC \times \max\{L_i\} \quad (8)$$

$$EC_{With} = \left(\sum_{i=1}^N T_i \times G_i \right) + DC \times \max\{G_i\} \quad (9)$$

$$DPP = \frac{-\ln \left(1 - \frac{C \times E_b \times r}{(EC_{With} - EC_{Without}) \times 365} \right)}{\ln(1 + r)} \quad (10)$$

$$NPV_{Without} = \sum_{y=0}^Y \frac{EC_{Without} \times 365}{(1 + r)^y} \quad (11)$$

$$NPV_{With} = \sum_{y=0}^Y \frac{EC_{With} \times 365}{(1 + r)^y} + C \times E_b \quad (12)$$

IV. RESULTS AND DISCUSSION

A. Optimal charging and discharging schedule

Fig. 1 and 2 show the load and grid import with battery capacities of 395kWh (base battery capacity) and 50kWh respectively. Note that from Eq. (5), the grid import G_i indicates the charge and discharge schedule of the BESS as the battery is charged at times when the grid import exceeds the load and is discharged when the grid import is less than the load. In Fig. 1, notice how the battery only charges in the off-peak period and only discharges in the peak period indicating that at this size, the main cost-minimizing discharge strategy is tariff arbitrage. However, with a capacity of only 50kWh, the operation of the BESS attempts to minimize the maximum demand charges by flattening the load.

B. Economic assessment of BESS

Considering the energy cost and the maximum demand charges, the BESS assessed has the potential to reduce the daily cost of electricity of the commercial facility by up to UGX87,300 as shown in Fig. 3. Focusing on the base BESS capacity which is the capacity that can fully power the load in the peak period, Table III details the energy cost and maximum demand charge with and without the BESS. While the maximum demand charge component of the electricity cost increased with the introduction of the BESS, the reduction in the energy cost was greater resulting in cost savings. With

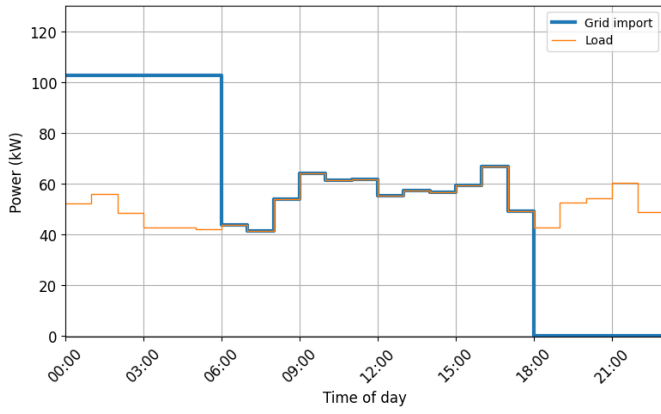


Fig. 1: Load profile and total grid import after installing a 395kWh BESS

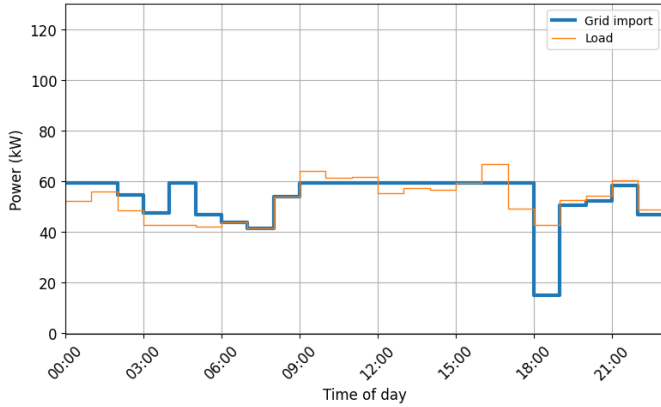


Fig. 2: Load profile and total grid import after installing a 50kWh BESS

these savings, the discounted payback of the BESS will be about 18 years for the battery unit cost of \$209/kWh, which is much higher than the expected lithium-ion BESS lifetime of 5 or 8 years. Table III also shows that the NPV of cash flows of the system with the BESS is about UGX 121 million higher than without the BESS over an 8-year period. Ignoring the time value of money ($r = 0$), the sum of the cashflows of the system with the BESS is still approximately UGX 59 million higher than in the scenario without the BESS. This implies that although the BESS is able to reduce the daily electricity cost, this reduction is not enough to offset the high capital expenditure of the system. In the next section, we show how sensitive the results of the analysis are to the most significant uncertain inputs.

TABLE III: Cost of electricity (UGX)

	Without BESS [1]	With BESS [2]	Difference [1] - [2]
Diurnal Energy cost	584,912.33	476,098.03	108,814.30
Diurnal Demand charges	46,154.88	71,096.86	(24,941.98)
Total	631,067.21	547,194.89	83,872.32
NPV (8 years lifetime, unit cost of \$209/kWh)	1.38B	1.50B	(121.45M)

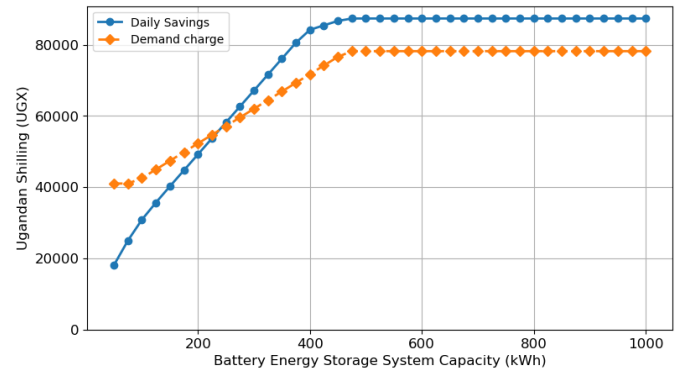


Fig. 3: Line chart showing how daily savings and the demand charges increase with increasing battery capacity until they plateau at around 475kWh.

C. Sensitivity analysis

1) *Impact of BESS capacity:* We explore the impact of different battery capacities on the NPV of the achieved savings. Figure 4 shows how the NPV of the savings over the lifetime of the BESS normalized by the BESS capacity varies with the BESS capacity. We assume a lifetime of 8 years and a discount rate of 7%. For battery capacity sizes beyond 75kWh, the NPV of savings per unit battery size (blue line) lies below the red dotted line (industry benchmark), indicating that the proposed solution is not cost-effective. This aligns with the result in [12] where smaller BESS were found to be the most economical.

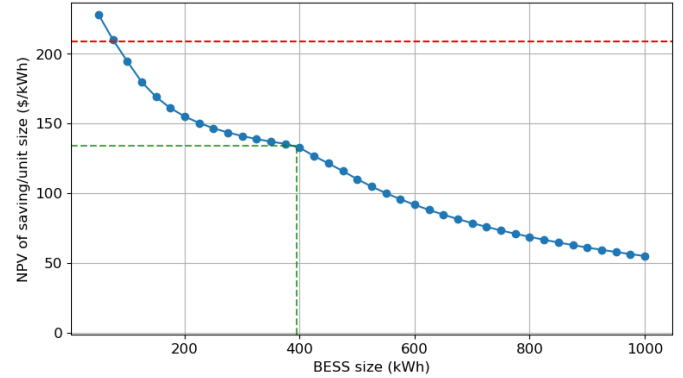


Fig. 4: Variation in NPV of savings with BESS capacity. The blue line shows the saving with the base capacity and the red dotted line represents the global industry benchmark cost of the BESS (\$209/kWh). Note that the industry benchmark does not include the cost of the power electronics devices that are required for the operation of the BESS.

2) *Second-life battery systems:* Given that the feasibility of the proposed solution depends heavily on the estimated daily savings and the unit cost of the BESS, we explore second-life battery systems which are typically reconditioned electric vehicle batteries that are considered no longer suitable for traction services although they still retain up to 80% of their capacity. These batteries have a lower unit cost than new batteries, albeit with a shorter lifespan. The cost of second-life

batteries has been estimated to range from \$40 to 160/kWh [22]. Substituting these costs into the expression in Eq. (10), the discounted payback period of the BESS for the base case varies from 2.1 years to 11.2 years with unit costs of \$40/kWh and \$160/kWh respectively. The unit cost of the base case that allows the system to break even at an expected lifetime of 7 years is \$113/kWh.

D. Implication of findings

Numerous research papers have underscored the necessity of diversifying revenue streams (also known as revenue/benefit stacking) to ensure the financial sustainability of BESS [8], [23]. In this study, only a single application, that is, tariff arbitrage was considered for the BESS. However, considering that firms in Uganda experience an average of 6.3 electrical outages every month with each outage lasting for an average of about 10 hours according to the 2014 World Bank Enterprise Survey [24], a BESS system installed here may also increase the reliability of electricity supply for commercial facilities. While this application was not considered and valued in this study, it has the potential to improve the financial viability of the BESS, especially if the backup systems currently used by the facilities have high operational costs.

V. CONCLUSIONS & FUTURE WORK

This study assessed the feasibility of behind-the-meter battery energy storage systems for tariff arbitrage for a commercial facility in Uganda. We found that although the introduction of the BESS may reduce the daily cost of electricity, the solution is not cost-effective because of the high capital expenditure required. The findings of this paper align with those of many other publications that have found that BESS is too expensive to be used for a single application and therefore add to the body of knowledge on the need for revenue stacking when assessing the economic feasibility of BESS for grid services. Other services such as improving power quality and improving reliability could be important applications that can improve the financial viability of the systems. In addition, this study also highlights the potential of second-life battery systems for grid services.

REFERENCES

- [1] C. Trimble, M. Kojima, I. Perez Arroyo, and F. Mohammadzadeh, "Financial Viability of Electricity Sectors in Sub-Saharan Africa; Quasi-Fiscal Deficits and Hidden Costs," World Bank, Washington, DC, Policy Research Working Paper 7788, 2016.
- [2] US. EIA, "Electric Power Monthly," US EIA, Report, Jan. 2023. [Online]. Available: <https://www.eia.gov/electricity/monthly/>
- [3] L. Maqelepo, N. Williams, and J. Taneja, "Rural electrification subsidy estimation: a spatial model development and case study," *Environmental Research: Infrastructure and Sustainability*, vol. 2, no. 4, p. 045009, Dec. 2022.
- [4] N. Rosenberg, "The role of electricity in industrial development," in *Studies on Science and the Innovation Process*. World Scientific, Aug. 2009, pp. 137–151.
- [5] A. Eberhard, V. Foster, C. Briceño-Garmendia, F. Ouedraogo, D. Camos, and M. Shkaratan, "Underpowered: The State of the Power Sector in Sub-Saharan Africa," World Bank, Washington, DC, Report, 2008. [Online]. Available: <http://hdl.handle.net/10986/7833>
- [6] M. Kojima and J. J. Han, "Electricity Tariffs for Nonresidential Customers in Sub-Saharan Africa," World Bank, Live Wire 2017/77, 2017.
- [7] T. Raji, L. Maqelepo, N. J. Williams, and A. Bett, "Money and Power: The Impact of Tariff Structures on Electricity Consumption in Solar Microgrids in Africa," in *2022 IEEE PES/IAS PowerAfrica*. Kigali, Rwanda: IEEE, Aug. 2022, pp. 1–5. [Online]. Available: <https://ieeexplore.ieee.org/document/9905330/>
- [8] W. Seward, M. Qadrdan, and N. Jenkins, "Revenue stacking for behind the meter battery storage in energy and ancillary services markets," *Electric Power Systems Research*, vol. 211, p. 108292, Oct. 2022.
- [9] K. Bradbury, L. Pratson, and D. Patiño-Echeverri, "Economic viability of energy storage systems based on price arbitrage potential in real-time U.S. electricity markets," *Applied Energy*, vol. 114, pp. 512–519, Feb. 2014.
- [10] M. Kloess, "Electric storage technologies for the future power system — an economic assessment," in *2012 9th International Conference on the European Energy Market*, 2012, pp. 1–8.
- [11] S. R. Deeba, R. Sharma, T. K. Saha, D. Chakraborty, and A. Thomas, "Evaluation of technical and financial benefits of battery-based energy storage systems in distribution networks," *IET Renewable Power Generation*, vol. 10, no. 8, pp. 1149–1160, Sep. 2016.
- [12] A. Odukumaiya, A. Abu-Heiba, and B. Bekker, "The value of behind-the-meter energy storage for buildings: A case study on a university building in South Africa," in *2018 IEEE PES/IAS PowerAfrica*, 2018, pp. 640–645.
- [13] T. Baloyi and S. Chowdhury, "Sizing and selection of battery energy storage system for time of use arbitrage in a commercial building in South Africa," in *2021 IEEE PES/IAS PowerAfrica*, 2021, pp. 1–5.
- [14] T. A. Nguyen and R. H. Byrne, "Maximizing the cost-savings for time-of-use and net-metering customers using behind-the-meter energy storage systems," in *2017 North American Power Symposium (NAPS)*. Morgantown, WV: IEEE, Sep. 2017, pp. 1–6.
- [15] GET.Invest, "Uganda: Captive Power Model Business Case: Solar Photovoltaic (PV) for Commercial and Industrial Facilities," Tech. Rep., 2019. [Online]. Available: https://www.get-invest.eu/wp-content/uploads/2020/11/GETInvest-Market-Insights_UGA_Captive_MBC-Facilities_2019.pdf
- [16] Umeme, "Electricity Retail Tariffs for Quarter One 2023," Umeme, Tech. Rep., Dec. 2022. [Online]. Available: https://www.umeme.co.ug/umeme_api/wp-content/uploads/2023/01/Q1_2023_Tariffs.pdf
- [17] Payway, "Umeme yaka rates," Payway, Tech. Rep., 2023. [Online]. Available: <https://paywaykiosk.com/umeme-yaka/>
- [18] W. E. Hart, J.-P. Watson, and D. L. Woodruff, "Pyomo: modeling and solving mathematical programs in python," *Mathematical Programming Computation*, vol. 3, no. 3, pp. 219–260, 2011.
- [19] HOMER Energy, "HOMER Energy," 2023. [Online]. Available: <https://www.homerenergy.com/>
- [20] C. Greacen, "Minigrid Costing and Innovation: Minigrids for Half a Billion People," Accra, Ghana, 2019. [Online]. Available: <https://atainsights.com/wp-content/uploads/2019/06/4.A.Chris-Greacen.World-Bank-consultant-notes.pdf>
- [21] USAID, "What are the key advances in mini-grid energy storage?" Dec. 2022. [Online]. Available: <https://www.usaid.gov/energy/mini-grids/emerging-tech/storage>
- [22] R. Madlener and A. Kirmas, "Economic Viability of Second Use Electric Vehicle Batteries for Energy Storage in Residential Applications," *Energy Procedia*, vol. 105, pp. 3806–3815, 2017. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S187661021730989X>
- [23] H. C. Hesse, M. Schimpe, D. Kucevic, and A. Jossen, "Lithium-ion battery storage for the grid—a review of stationary battery storage system design tailored for applications in modern power grids," *Energies*, vol. 10, no. 12, 2017. [Online]. Available: <https://www.mdpi.com/1996-1073/10/12/2107>
- [24] World Bank, "Enterprise Surveys," World Bank, Tech. Rep., 2014. [Online]. Available: www.enterprisesurveys.org